(SHORTLEX) CONE TYPES IN A GROUP WITH FFTP (YAGO HAD A BETTER SUGESITON OF TITLE)

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Abstract. We show that a finitely generated group $G = \langle \Sigma \rangle$ satisfying the falsification by fellow traveller property also satisfies the shortlex falsification by fellow traveller property for any choice of total ordering on the generating set Σ, and that the corresponding language of shortlex representatives $\mathrm{SL}_G \subset (\Sigma \cup \Sigma^{-1})^*$ is a regular language.

1. Introduction

The *falsification by fellow traveller property* (FFTP) was introduced by Neumann and Shapiro in [11], who were inspired by the ideas of Cannon in [3]. Informally, a graph has the FFTP if every non-geodesic path in the graph admits a shorter path, with the same endpoints, that remains uniformly close to the first one. We say that a group G with finite generating set Σ has the FFTP with respect to S if the Cayley graph Cay(G, Σ) has it.

From the work of Neumann and Shapiro, one can see that if a group G has the FFTP with respect to some generating set Σ, then the language of geodesics over this generating set is regular. In addition to the results in the grounding works of Cannon, Neumann and Shapiro, the falsification by fellow traveller property has been the topic of many research articles. Indeed, it has been shown that groups with FFTP are of type F3 and their Dehn function is quadratic [6], that they have rational growth, and that having a regular set of geodesics does not imply having FFTP [5]. Moreover, there are many works providing new examples of families of groups that satisfy FFTP: Dyer groups [10], Coxeter groups [12], dihedral Artin groups [4], Garside groups [8], Artin groups of large type [9] and relatively hyperbolic groups [1].

In the literature, there exist variations of the FFTP for graphs *relative to a subset of paths in the graph* (see [2]). In the definition of these variations, the role of geodesics is played by the chosen subset of paths in the graph. An example of this is the *shortlex FFTP*, where the role of geodesics is played by shortlex paths in a graph whose edges are labelled by letters in an alphabet A.

For a fixed total order in the letters of an alphabet A, it is possible to define a well-ordering on the set of words over the alphabet A, A^*, which is called the *shortlex ordering* of A^* (see Definition 2.7). If G is a group with generating set Σ, we can consider the shortlex ordering on $(\Sigma \cup \Sigma^{-1})^*$ and take the unique minimal representative of each element of the group with respect to this ordering. We say that a group G together with a finite generating set Σ satisfies the *shortlex falsification by fellow traveller property* (SL-FFTP) if there exists a global constant K such that, for every word that is not a minimal shortlex representative, we can find another representative of the same group element which is smaller in the shortlex ordering and K-fellow travels with it.

The main goal of this paper is to prove the following theorem:

Theorem A. *Let G be a group with finite generating set Σ satisfying* FFTP*, then* SL_G *is a regular language.*

The strategy to prove this result is to mimic the classical argument by Neumann and Shapiro for showing that having FFTP implies that the set of geodesics is regular. Essentially, we will define a shortlex analog of the cone of an element (see Definition 3.2) and conclude that there are finitely many of them, which allows us to build a finite state automaton recognising the language of minimal shortlex representatives of elements of the group.

The rest of this document is structured as follows: Section 2 provides preeliminary definitions regarding regularity of languages, the falsification by fellow-traveller property and the shortlex version of this same property, and Section 3 contains the main results.

2. Preeliminary definitions

Definition 2.1 (Finite state automaton)**.** A *finite state automaton* (FSA) is a tuple $\mathcal{A} = (\Sigma, V, E, V_F, v_0)$ where Σ is a finite set of letters (the *alphabet*), V is the set of vertices or *states*, E is the set of Σ-labelled edges or *transitions* of the form $(v, s, w) \in V \times \Sigma \times V$, $v_0 \in V$ is the *initial state* and $V_F \subset V$ is the set of *accepting states* of the automaton.

We say that $(v, s, w) \in E$ is an *s-labelled* transition.

Definition 2.2 (Language accepted by FSA, regular language)**.** To each finite state automaton $\mathcal{A}$ we can assign the set $L(\mathcal{A}) \subset \Sigma^*$ of words *accepted* by the automaton, which are the words given by the concatenation of edge labels of paths in $\mathcal{A}$ that start in v_0 and end in one of the accepting states; we call $L(\mathcal{A})$ the *language* accepted by $\mathcal{A}$. We say that a subset L of Σ^* is *regular* if it is the language $L(\mathcal{A}_L)$ accepted by a finite state automaton $\mathcal{A}_L$.

For further definitions related to finite state automata and regular languages we refer the reader to [7, §1.1 and §1.2].

Given a graph X, a *(combinatorial) path* p is a sequence $v_0 e_1 v_1 \dots e_n v_n$ where $v_0, \dots, v_n$ are vertices in X and, for $i = 1, \dots, n$, e_i is an edge between v_i and v_{i+1}. The *length* of p is the number n, and the vertices v_0, v_n are called the *endpoints* of the path. The path p is *geodesic* if its length is minimal among the lengths of all the paths in X with endpoints v_0,v_n.

Definition 2.3 (Fellow traveller for graphs)**.** For $K \geq 0$, we say that two paths $p = v_1 \dots v_n$, $q = u_1 \dots u_m$ in a graph X *asynchronously K-fellow travel* if there is a non-decreasing, proper map $f\colon \mathbb{N} \to \mathbb{N}$ satisfying that the vertices $v_i, u_{f(i)}$ are at distance at most K in the graph for all $i \geq 0$. We say that p and q *synchronously K-fellow travel* if v_i, u_i are at distance at most K for all $i \geq 0$.

Definition 2.4 (FFTP for graphs)**.** A graph X has the *(asynchronous) falsification by fellow traveller property* if there is a constant K such that for every non-geodesic path p there exists a path q with the same endpoints as p, that is shorter than p and that K-fellow travels with it.

These concepts are translated to a group using its Cayley graph with respect to a given set of generators. For any group G with finite generating set Σ, its elements are represented by words $w \in (S \cup S^{-1})^*$. Each such word w corresponds to a combinatorial path in the Cayley graph of G with respect to S starting at a fixed basepoint, take the basepoint to be the vertex corresponding to the identity 1_G.

The length of such a word w, which coincides with the length of the corresponding path, will be denoted by $|w|$. The length of the shortest $v \in (\Sigma \cup \Sigma^{-1})^*$ such that $v =_G w$ is denoted by $|w|_G$. We say w is geodesic if its corresponding path is geodesic or, equivalently, if $|w| = |w|_G$.

Definition 2.5 (Fellow traveller for groups)**.** Given a group G with finite generating set Σ and a constant $K \geq 0$, we say that two words $w, v \in (\Sigma \cup \Sigma^{-1})^*$ *asynchronously K-fellow travel* if the corresponding paths p, q asynchronously K-fellow travel in the Cayley graph of G with respect to Σ. We say that w and v *synchronously K-fellow travel* if p and q synchronously fellow travel in the Cayley graph.

The falsification by fellow traveller property for groups was introduced by Neumann and Shapiro in [11]. Two notions appear in the literature: the synchronous and the asynchronous falsification by fellow traveller property. In [6], Elder showed that these two notions are actually equivalent, so we choose to work with the asynchronous one.

Definition 2.6 (FFTP for groups)**.** A group G with generating set Σ is said to have the *falsification by fellow traveller property* (FFTP) *with respect to* Σ if the Cayley graph of G with respect to Σ has the asynchronous falsification by fellow traveller property.

Next, we introduce the shortlex ordering for words over an alphabet with respect to a given total ordering of the alphabet.

LAM: Como en el otro paper, cambio X por S para no marear con que X es un grafo.

Definition 2.7 (Shortlex ordering)**.** Let $(S, <)$ be a totally ordered alphabet. We will consider two different order relations in S^*:

- The *lexicographical order*: $w_1, w_2 \in S^*$, $w_1 <_{\text{lex}} w_2$ if and only if w_1 is a strict prefix of w_2 or the letter in the first position in which w_1 and w_2 differ is smaller for w_1.
- The *shortlex order*: for $w_1, w_2 \in S^*$, $w_1 <_{\text{SL}} w_2$ if and only if w_1 is shorter than w_2 or they have the same length and $w_1 <_{\text{lex}} w_2$.

Notation 2.8. The shortlex order is a well-ordering. In particular, if G is a group generated by Σ and we consider the shortlex order on $(\Sigma \cup \Sigma^{-1})^*$ with respect to some ordering of $\Sigma \cup \Sigma^{-1}$, there is a well-defined minimal representative for each element g of the group G under this order, which will be denoted $\text{SL}_{(\Sigma,<)}(g)$ or simply $\text{SL}(g)$ if $(\Sigma, <)$ is understood by the context. We will denote by $\text{SL}_{(G,\Sigma,<)}$ the language of these minimal representatives of elements of G, and write SL_G for short if $(G, \Sigma, <)$ is understood.

As mentioned before, variations of the FFTP arise by taking other subsets of words to play the role of geodesic words in the definition. We will use one of these variations, the shortlex version of the falsification by fellow traveller property, which is introduced next.

Definition 2.9 (SL-FFTP)**.** Let G be a group with generating set Σ, and fix $<$ a total ordering on Σ. The group G is said to have the *shortlex falsification by fellow* *traveller property* (SL-FFTP) *with respect to* $(\Sigma, <)$ if there exists a constant $K \geq 0$ such that, for any $w \notin \mathrm{SL}_G$, there exists a word $v \in (\Sigma \cup \Sigma^{-1})^*$ that K-fellow-travels with w and satisfies that $w =_G v$ and $v <_{\mathrm{SL}} w$.

3. FFTP implies SL_G is regular

The results in this section allow us to see that groups satisfying FFTP also satisfy a shortlex version of this property and that the set of minimal shortlex representatives is a regular language.

From now on, G will always be a group with finite generating set Σ. We fix a total order in $\Sigma \cup \Sigma^{-1}$. Every time we use the shortlex ordering in $(\Sigma \cup \Sigma^{-1})^*$, we will assume that we are referring to the shortlex order induced by this fixed total order.

Lemma 3.1. *If G satisfies* FFTP *with respect to Σ, then G satisfies* SL-FFTP *with respect to Σ.*

Proof. Let K be the FFTP constant of G with respect to Σ. Let $w \notin \mathrm{SL}_G$. If w is not geodesic, there exists a shorter word $v \in (\Sigma \cup \Sigma^{-1})^*$ that K-fellow travels with w and represents the same element of G, since G has FFTP. In particular, $v <_{\mathrm{SL}} w$, so SL-FFTP is satisfied in this case. Suppose now that w is geodesic, then there exists $u \in (\Sigma \cup \Sigma^{-1})^*$ with $|u| = |w|$ admitting a factorization as the following: $u = u_1 x u_2$ and $w = w_1 y w_2$, with $u_i, w_i \in (\Sigma \cup \Sigma^{-1})^*$, $x, y \in \Sigma \cup \Sigma^{-1}$, $u_1 = w_1$ and $x < y$. The word $x^{-1} y w_2$ is not geodesic (since it represents the same element as u_2 and it has length $|u_2| + 2$) so, by FFTP, there exists a word v that K-fellow travels with $x^{-1} y w_2$, represents the same element of G and satisfies $|u_2| \leq |v| < |x^{-1} y w_2| = |u_2| + 2$. In particular, $|v| = |u_2|$ or $|v| = |u_2| + 1$. By FFTP, we may assume that $|v| = |u_2|$. Then, $w_1 x v$ K-fellow travels $w = w_1 y w_2$, $w_1 x v <_{\mathrm{SL}} w$ and $w =_G w_1 x v$, so G has SL-FFTP. □

Definition 3.2. Let G be a group with finite generating set Σ and let g be an element of G. The SL-cone of g is defined as

$$\mathrm{C}_{\mathrm{SL}}(g) := \{w \in (\Sigma \cup \Sigma^{-1})^* : \mathrm{SL}(g) w \in \mathrm{SL}_G\}.$$

Lemma 3.3. *Let G be a group with finite generating set Σ. If G satisfies the* FFTP *with respect to Σ, then G has a finite number of* SL*-cone types.*

Proof. Let K be the maximum of the FFTP and the SL-FFTP constants (both properties hold by Lemma 3.1). Denote by $B_K(1_G)$ the ball of radius K and center 1_G in the Cayley graph of G with respect to Σ. For $g \in G$ and $h \in B_K(1_G)$, we define

$$\Delta_g(h) := |gh|_G - |g|_G$$

$$L_g(h) := \begin{cases} 0, & \text{if } \mathrm{SL}(gh) <_{\mathrm{lex}} \mathrm{SL}(g), \\ 1, & \text{if not.} \end{cases}$$

$$P_g(h) := \begin{cases} 0, & \text{if } \mathrm{SL}(gh) \text{ is not a prefix of } \mathrm{SL}(g), \\ 1, & \text{if } \mathrm{SL}(gh) \text{ is a prefix of } \mathrm{SL}(g). \end{cases}$$

$$T_g(h) := \begin{cases} s, & \text{if } P_g(h) = 1 \text{ and } \mathrm{SL}(g) = \mathrm{SL}(gh)s, \\ \varepsilon, & \text{if } P_g(h) = 0. \end{cases}$$

Notice that $\Delta_g(h) \in [-K, K]$ and $T_g(h)$ is a word of length at most K for all $g \in G, h \in B_K(1_G)$. Thus, if we denote by $[(\Sigma \cup \Sigma^{-1})^*]_{\leq K}$ the set of words in $(\Sigma \cup \Sigma^{-1})^*$ of length at most K, we may define:

$$\begin{array}{rcl} \mathcal{C}_g : B_K(1_G) & \longrightarrow & [-K, K] \times \{0,1\}^2 \times [(\Sigma \cup \Sigma^{-1})^*]_{\leq K} \\ h & \longmapsto & (\Delta_g(h), L_g(h), P_g(h), T_g(h)). \end{array}$$

Observe that $\{\mathcal{C}_g\}_{g \in G}$ is a finite set. So, if we take $g, g' \in G$ with $\mathcal{C}_g(h) = \mathcal{C}_{g'}(h)$ for all $h \in B_K(1_G)$, it is enough to show that $\mathrm{C}_{\mathrm{SL}}(g) = \mathrm{C}_{\mathrm{SL}}(g')$ in order to prove that the number of SL-cones of G is finite.

Assume that $g, g' \in G$ satisfy that $\mathcal{C}_g(h) = \mathcal{C}_{g'}(h)$ for all $h \in B_K(1_G)$. Suppose, for a contradiction, that there exists $w \in \mathrm{C}_{\mathrm{SL}}(g) - \mathrm{C}_{\mathrm{SL}}(g')$. Since G satisfies SL-FFTP and $\mathrm{SL}(g')w$ is not geodesic, there is a word v that K-fellow travels with $\mathrm{SL}(g')w$ and such that $v <_{\mathrm{SL}} \mathrm{SL}(g')w$ and $v =_G \mathrm{SL}(g')w$ (by FFTP, we may assume that $|v| < |\mathrm{SL}(g')w|$ if $\mathrm{SL}(g')w$ is not geodesic). In particular, there is $h \in B_K(1_G)$ so that $g'h$ is a vertex of the path labelled by v starting at 1_G. Let u be the suffix of v such that $vu^{-1} =_G g'h$, that is, the label of the final subpath of the path labelled by v starting at $g'h$.

- **Case 1:** Suppose $\mathrm{SL}(g')w$ is not geodesic, so $|v| < |\mathrm{SL}(g')w|$. Then, we can argue as in the standard proof (the one to show that having FFTP implies that there is a finite number of cones) in order to obtain

$$\begin{array}{rcl} |\mathrm{SL}(gh)u| & = & |\mathrm{SL}(gh)| + |u| - |\mathrm{SL}(g)| + |\mathrm{SL}(g)| \\ & \overset{\Delta_g(h)=\Delta_{g'}(h)}{=} & |\mathrm{SL}(g'h)| + |u| - |\mathrm{SL}(g')| + |\mathrm{SL}(g)| \\ & \leq & |v| - |\mathrm{SL}(g')| + |\mathrm{SL}(g)| \\ & < & |\mathrm{SL}(g')| + |w| - |\mathrm{SL}(g')| + |\mathrm{SL}(g)| = |\mathrm{SL}(g)w|, \end{array}$$

  which contradicts the fact that $\mathrm{SL}(g)w \in \mathrm{SL}_G$.
- **Case 2:** Suppose $\mathrm{SL}(g')w$ is geodesic, so $|v| = |\mathrm{SL}(g')w|$ and $v <_{\mathrm{lex}} \mathrm{SL}(g')w$. In this case, given that the words have the same length, v cannot be a strict prefix of $\mathrm{SL}(g')w$, so it is possible to factorize as follows: $v = v_1xv_2$ and $\mathrm{SL}(g')w = t_1yt_2$ with $v_1 = t_1$ and $x < y$. Moreover,

$$\begin{array}{rcl} |\mathrm{SL}(gh)u| & = & |\mathrm{SL}(gh)| + |u| - |\mathrm{SL}(g)| + |\mathrm{SL}(g)| \\ & \overset{\Delta_g(h)=\Delta_{g'}(h)}{=} & |\mathrm{SL}(g'h)| + |u| - |\mathrm{SL}(g')| + |\mathrm{SL}(g)| \\ & = & |v| - |\mathrm{SL}(g')| + |\mathrm{SL}(g)| \\ & = & |\mathrm{SL}(g')| + |w| - |\mathrm{SL}(g')| + |\mathrm{SL}(g)| = |\mathrm{SL}(g)w|. \end{array}$$

  - **Case 2.1:** Suppose $\mathrm{SL}(g')$ is a prefix (not necessarily strict) of t_1. In this case, we may write $w = w_1yw_2$ and we have that $\mathrm{SL}(g)w_1xv_2 <_{\mathrm{SL}} \mathrm{SL}(g)w$, contradicting $\mathrm{SL}(g)w \in \mathrm{SL}_G$.
  - **Case 2.2:** Suppose t_1y is a prefix of $\mathrm{SL}(g')$.
    - **Case 2.2.A:** (See Fig. 1). Suppose that $g'h$ is a vertex of the path labelled by v_1. Since we are also assuming that t_1y is prefix of $\mathrm{SL}(g')$ and $v_1 = t_1$, it is clear that $\mathrm{SL}(g'h)$ is a prefix of $\mathrm{SL}(g')$. Now, because $P_g(h) = P_{g'}(h)$ and $T_g(h) = T_{g'}(h)$, there exists a word s such that $\mathrm{SL}(g') = \mathrm{SL}(g'h)s$ and $\mathrm{SL}(g) = \mathrm{SL}(gh)s$. As $g'h$ is a vertex of the path labelled by v_1, we can factorise $s = s_1ys_2$ and $u = u_1xu_2$ with $u_1 = s_1$. Therefore, $\mathrm{SL}(gh)u = \mathrm{SL}(gh)s_1xu_2$ is smaller with respect to the shortlex

ordering than $\mathrm{SL}(g)w = \mathrm{SL}(gh)s_1ys_2$. This gives a contradiction because $\mathrm{SL}(g)w =_G \mathrm{SL}(gh)u$ and $\mathrm{SL}(g)w$ is supposed to be geodesic.

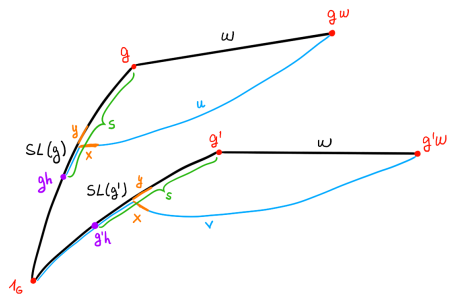

FIG. 1. Case 2.2.A in the proof of Lemma 3.3.

* **Case 2.2.B:** (See Fig. 2). Suppose now that $g'h$ is a vertex of the path labelled by v_2 and also a vertex of the path labelled by $\mathrm{SL}(g')$. This implies that $\mathrm{SL}(g'h)$ is a prefix of $\mathrm{SL}(g')$ and, because v is geodesic, t_1y must be a prefix of $\mathrm{SL}(g'h)$. Given that $x < y$, if we denote by $v_{g'h}$ the suffix of v labelling a path from 1_G to $g'h$, we get that $v_{g'h} <_{\mathrm{SL}} \mathrm{SL}(g'h)$, which is a contradiction.

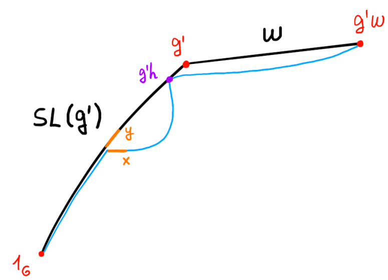

FIG. 2. Case 2.2.B in the proof of Lemma 3.3.

* **Case 2.2.C:** (See Fig. 3). Suppose now that $g'h$ is a vertex of the path labelled by v_2, but not a vertex of the path labelled by $\mathrm{SL}(g')$. Denote by $v_{g'h}$ the subpath of the path labelled by v that goes from 1_G to $g'h$. Since $\mathrm{SL}(g'h) \leq_{\mathrm{SL}} v_{g'h}$ and both words are geodesic, we have that $\mathrm{SL}(g'h) \leq_{\mathrm{lex}} v_{g'h}$. Moreover, since $x < y$, $v_{g'h} <_{\mathrm{lex}} \mathrm{SL}(g')$. By transitivity, it is clear that $\mathrm{SL}(g'h) <_{\mathrm{lex}} \mathrm{SL}(g')$, and using that $L_g(h) = L_{g'}(h)$, we get that $\mathrm{SL}(gh) <_{\mathrm{lex}} \mathrm{SL}(g)$. Observe that, because we are assuming that $g'h$ is not a vertex of the path labelled by $\mathrm{SL}(g')$, $\mathrm{SL}(g'h)$ is not a prefix of $\mathrm{SL}(g')$ and, because $P_g(h) = P_{g'}(h)$, we also have that $\mathrm{SL}(gh)$ is not a prefix of $\mathrm{SL}(g)$. As a consequence, there we can factorise $\mathrm{SL}(g) = w_1zw_2$ and $\mathrm{SL}(gh) = w_1tw_3$ with w_i

words on the generators and z and t generators satisfying that $t < z$. And finally, since $|\mathrm{SL}(gh)u| = |\mathrm{SL}(g)w|$, we conclude that $\mathrm{SL}(gh)u <_{\mathrm{SL}} \mathrm{SL}(g)w$, a contradiction.

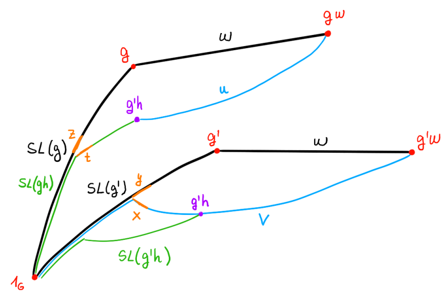

Fig. 3. Case 2.2.C in the proof of Lemma 3.3.

$\square$

Theorem 3.4. *Let G be a group with finite generating set Σ satisfying* FFTP*, then* SL_G *is a regular language.*

Proof. A finite automaton recognizing SL_G can be built as follows. Take the SL-cones of G to be the states of the automaton (there will be a finite number of them by Lemma 3.3), all states are final and there is a unique initial state: the SL-cone of 1_G. For each $s \in \Sigma \cup \Sigma^{-1}$ and $g \in G$, there is a transition from $\mathrm{C}_{\mathrm{SL}}(g)$ to $\mathrm{C}_{\mathrm{SL}}(gs)$ if and only if $s \in \mathrm{C}_{\mathrm{SL}}(g)$. $\square$

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